How to Build a Rapid Sand Filter

WHAT IS A RAPID SAND FILTER?

A rapid sand filter uses different types of soil, such as rock, gravel, and sand, to filter out particles and debris in the water. This helps improve the turbidity or clarity of the water. It uses gravity to purify the water and this process is typically used in drinking water facilities.

MATERIALS

- Plastic 1-liter bottle
- 2 cups
- Soil media types: rock, gravel, and sand.
- Filter paper: paper towel, old sock, or coffee filter
- Rubber band to wrap around the bottle
- Soil with debris: twigs, leaves, etc.
- Scissors
- Gloves
- Access to water faucet

INSTRUCTIONS

1. Retrieve all materials needed. Be sure to place gloves on.
2. Prepare your "dirty" water used in this experiment by adding your soil with debris into your 1 cup of water.
3. Take your 1-liter bottle and cut it in half. Throw away the bottom half and keep the top half which has the bottle cap on it.
4. Using the top half (remove the cap), take your filter paper and place it on top of the opening. Place a rubber band around the filter paper and the bottle.
5. Take the top half of the bottle and place it upside down so that the bottle opening is facing downward on the surface.
6. You have the choice to use one, two, or three soil types:
    a. Single type: Using only the sand, pack the sand inside of the filter bottle
    b. Two types: Using the sand and gravel/pebbles, pack the sand in the bottle first. Then place the gravel/pebbles on top of the sand.
    c. Three types: Using the sand, gravel, and large rocks, pack the sand inside the bottle, followed by the gravel, and followed by the larger rocks.
7. Taking your second cup, place the 1-liter bottle onto the cup with the opening face down inside the cup.
8. Using your choice of soil media types, take your prepared "dirty water" and pour into your rapid sand filter. Wait for your sample to filter through.
9. Your cup should look much clearer than your prepared "dirty" sample water. Observe the difference.

